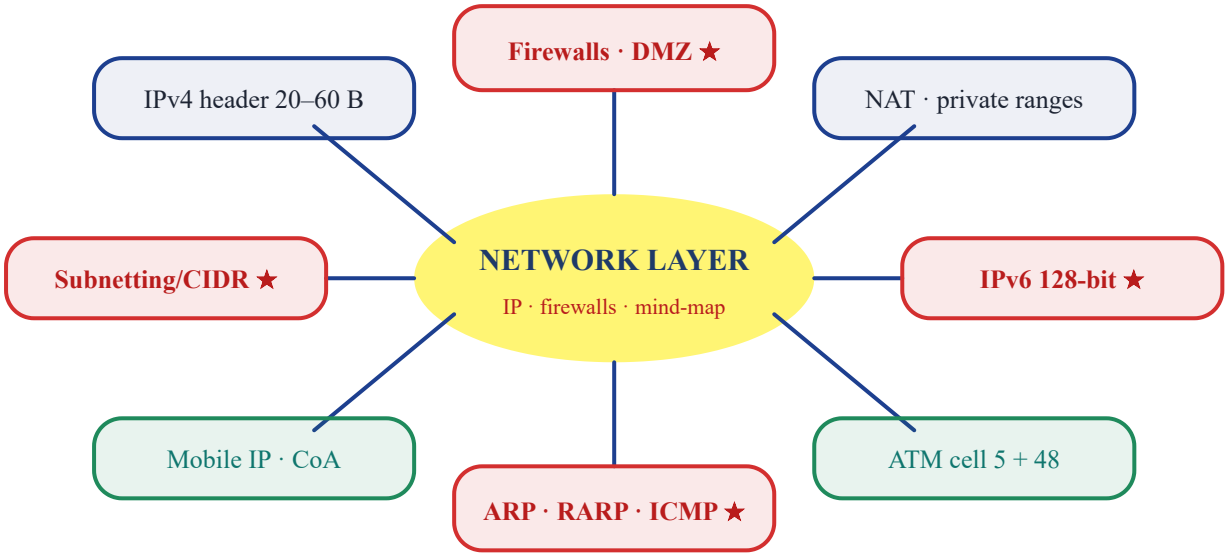
Firewalls, IPv4, IPv6, ARP, Mobile IP & ATM

Firewalls · IPv4 · subnetting · NAT · IPv6 · ARP · Mobile IP · ATM: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = applied / less frequent

Syllabus coverage

"Computer Networking and Security: … Fragmentation; **Firewalls**; **IPv4, IPv6, ARP, RARP, Mobile IP, Network layer of ATM**; Transport protocols …". Covers the firewall at the network border, IPv4/IPv6 packets and addresses, the helpers (ARP, RARP, ICMP, NAT), mobility and the ATM network layer. Fragmentation arithmetic and routing: see the Routing notes.

High-yield list

① firewall types · DMZ
② IHL × 4 · TTL
③ private ranges
④ hosts = $2^h - 2$
⑤ IPv6 "::" once
⑥ ARP: IP → MAC
⑦ HA · FA · CoA
⑧ ATM 53 = 5 + 48

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

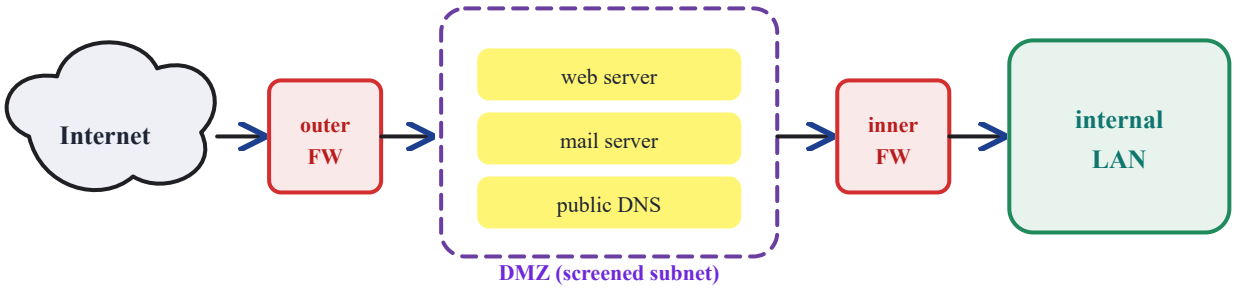
Contents

✓ Firewalls, DMZ, ACLs …… p. 2
✓ IPv4 header …… p. 3
✓ Classes, special ranges, NAT …… p. 4
✓ Subnetting, CIDR, VLSM …… p. 5
✓ IPv6 addresses …… p. 6
✓ IPv6 header & transition …… p. 7
✓ ARP, RARP, ICMP …… p. 8
✓ Mobile IP & ATM network layer … p. 9
✓ Night-before sheet …… p. 10
✓ Practice MCQs + written …… p. 11

1 Firewalls

Definition: A firewall is hardware or software at a network boundary that **permits or blocks traffic according to a rule set** (security policy). **All** traffic between inside and outside must pass through it. The safe policy is **default deny** : whatever is not explicitly allowed is blocked.

Type	Layer	Decides on	+ / −
Packet filter (stateless)	3–4	IP addresses, ports, protocol, TCP flags; each packet alone	fast, cheap (router ACL) / no state, fooled by spoofing
Stateful inspection	3–4	packet + **state table** of open connections	replies pass automatically / table can be flooded
Circuit-level gateway	5	validates the session set-up, then relays (SOCKS)	hides inside hosts / no content check
Application gateway (proxy)	7	full content of one protocol (HTTP, FTP, SMTP), user login, logs	strongest control / slow, one proxy per service
NGFW (next-generation)	3–7	deep packet inspection, IPS, application and user identity	all in one box / cost, latency



screened subnet: public servers sit between two firewalls; the inside LAN is never reached directly

Architectures

- **Screening router**: packet filtering only
- **Dual-homed host**: **bastion host**, two NICs, forwarding off
- **Screened host**: filtering router + bastion host
- **Screened subnet (DMZ)**: two firewalls or one three-legged firewall
- **Host-based** (Windows Defender Firewall, iptables) vs **network-based**

Memory trick: The **packet filter reads the envelope**, the **stateful** firewall **remembers the conversation**, the **proxy opens the letter**.

ACL: first match wins

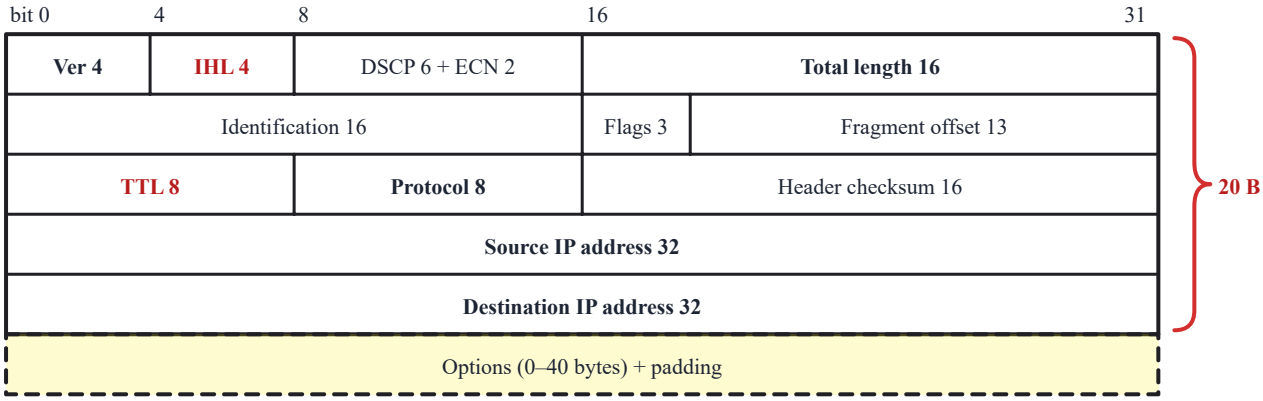
```
permit tcp any host 203.0.113.10 eq 443
permit tcp any host 203.0.113.25 eq 25
permit tcp any 10.0.0.0/8 established
deny   ip  any any   # implicit anyway
```

A broad **permit** above a narrow **deny** makes the deny useless.

Exam trap: A firewall cannot stop traffic that **bypasses** it (insider, USB stick, rogue Wi-Fi) or malware hidden in allowed, encrypted traffic. **IDS** only detects and alerts; **IPS** sits inline and blocks.

2 The IPv4 Datagram Header

IPv4: (RFC 791, **1981**): a **connectionless, best-effort** network-layer protocol: no acknowledgements, no ordering, no retransmission (TCP adds those). Datagram = header (**20–60 bytes**) + data; maximum total size **65,535 bytes**.



5 rows × 32 bits = 160 bits = 20 bytes fixed part

Field	Bits	What to remember
Version	4	value 4 (0100); IPv6 packets carry 6
IHL (header length)	4	counts **32-bit words**: 5–15 → **20–60 bytes**
DSCP + ECN (old ToS)	6 + 2	QoS class (DiffServ) and congestion marking
Total length	16	header + data in bytes, max $2^{16} - 1 = 65{,}535$
Identification, Flags, Fragment offset	16, 3, 13	fragmentation: flags = reserved, **DF**, **MF**; offset in 8-byte units
TTL (time to live)	8	−1 at every router; at 0 the packet is dropped and ICMP **Time Exceeded** is sent
Protocol	8	**1 ICMP · 6 TCP · 17 UDP** · 2 IGMP · 41 IPv6 · 50 ESP · 51 AH · 89 OSPF
Header checksum	16	covers the **header only**; recomputed at every hop (TTL changes)
Source, Destination	32 + 32	IPv4 addresses; unchanged end to end (except by NAT)

Worked example ✍️: First byte **0x45** → version 4, IHL 5 → header 5 × 4 = **20 bytes**. If IHL = 8, header = 32 bytes, so 12 bytes are options. With total length 1500 and IHL 5, data = 1500 − 20 = **1480 bytes**. Largest possible data in one datagram: 65,535 − 20 = 65,515 bytes.

Memory trick: "**IHL × 4, offset × 8**": header length counts 4-byte words, fragment offset counts 8-byte blocks. Protocol numbers: "**1 ping, 6 TCP, 17 UDP**".

Exam trap: The IPv4 checksum does **not** protect the data (TCP/UDP checksums do). TTL is in practice a **hop count**, not seconds. Offset/MF arithmetic: see the Fragmentation sheet of the Routing notes.

3 IPv4 Addresses, Classes & Special Ranges

Address: 32 bits, four decimal octets; **$2^{32} \approx 4.3$ billion** addresses, each = **network ID + host ID**. Classful (1981) split by the first bits; **CIDR** (1993) allows any prefix /n.

Class (bits)	First octet	Default mask	Networks	Hosts / network
A (0)	**1–126**	/8 = 255.0.0.0	$2^7 - 2 = 126$	$2^{24} - 2 = 16{,}777{,}214$
B (10)	**128–191**	/16 = 255.255.0.0	$2^{14} = 16{,}384$	$2^{16} - 2 = 65{,}534$
C (110)	**192–223**	/24 = 255.255.255.0	$2^{21} = 2{,}097{,}152$	$2^8 - 2 = 254$
D (1110)	224–239	multicast (no network/host split)		
E (1111)	240–255	reserved / experimental		

Special address	Meaning
0.0.0.0	this host / default route
127.0.0.0/8	loopback (127.0.0.1)
255.255.255.255	limited broadcast
host bits all 0 / 1	network / directed bcast
169.254.0.0/16	link-local (APIPA)
100.64.0.0/10	carrier-grade NAT

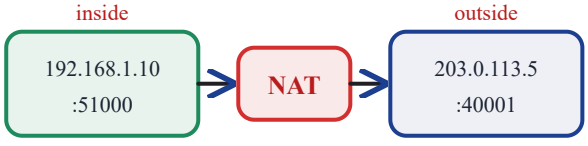
Private (RFC 1918):
10.0.0.0/8 · 10.x.x.x
172.16.0.0/12
172.16.x.x – 172.31.x.x
192.168.0.0/16 · 192.168.x.x
not routed on the Internet

Exam trap: 172.**32**.x.x and 192.**169**.x.x are **public**. 127 is kept for loopback, so usable class A first octets are 1–126, not 0–127.

4 NAT (Network Address Translation)

The border router **rewrites private source addresses to a public one** (RFC 3022) and reverses it for replies.

- **Static NAT**: 1 private ↔ 1 public (to publish a server)
- **Dynamic NAT**: a shared pool of public addresses



one table row per connection

PAT rewrites address and port

– **PAT / NAPT** ("overload"): many hosts → **one** public IP, told apart by **port numbers**

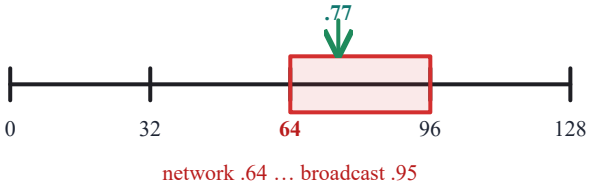
Memory trick: Private = "**10 · 172.16 to 31 · 192.168**": one /8, sixteen class-B blocks, 256 class-C blocks.

Exam trap: NAT is **not** a firewall, though it drops unsolicited inbound packets. It breaks end-to-end addressing: IPsec AH, active FTP and peer-to-peer need ALGs or NAT traversal.

5 Subnetting, CIDR & VLSM

The formulas: prefix /n → host bits **h = 32 − n** · addresses per subnet = 2^h · usable hosts = $2^h - 2$ (network and broadcast removed) · borrowing s bits gives 2^s subnets · **block size = 256 − mask octet** · network = largest multiple of the block ≤ the address · broadcast = next network − 1

Prefix	Mask	Block	Hosts
/24	255.255.255.0	256	254
/25	…255.128	128	126
/26	…255.192	64	62
/27	…255.224	32	30
/28	…255.240	16	14
/29	…255.248	8	6
/30	…255.252	4	**2**
/20	255.255.240.0	16 (3rd)	4,094
/22	255.255.252.0	4 (3rd)	1,022



/27: count in blocks of 32

/31 (RFC 3021) is used on point-to-point links: 2 addresses, both usable. /32 = a single host route.

Memory trick: "**Magic number** = 256 − mask octet". Count up in magic numbers; the address falls in one block.

Example 1 ✍️: 192.168.10.130/26 → block 64: blocks start .0, .64, **.128**, .192 → network **192.168.10.128**, hosts .129–.190, broadcast **.191**, $2^6 - 2 = 62$ hosts.

Example 2 ✍️: 172.16.45.200/20 → mask 255.255.240.0, block 16 in the **third** octet: 32 ≤ 45 < 48 → network **172.16.32.0**, broadcast 172.16.47.255, hosts 172.16.32.1–172.16.47.254 ($2^{12} - 2 = 4{,}094$).

Example 3: VLSM ✍️: Split 192.168.1.0/24 for LAN A (100 hosts), LAN B (50 hosts) and a router link (2). **Largest first:**

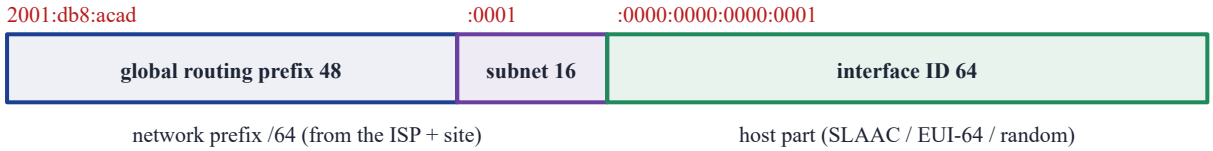
1. A needs $2^7 - 2 = 126 \geq 100$ → /25: **192.168.1.0/25** (.1–.126, bcast .127)
2. B needs $2^6 - 2 = 62 \geq 50$ → /26: **192.168.1.128/26** (.129–.190, bcast .191)
3. Link needs 2 → /30: **192.168.1.192/30** (.193, .194); .196–.255 is left free

Example 4: supernetting (route aggregation) ✍️: 200.10.0.0/24, 200.10.1.0/24, 200.10.2.0/24, 200.10.3.0/24: third octets 0–3 = 00000**00**…00000**11** share 16 + 6 = 22 bits → one route **200.10.0.0/22**. Rule: a power-of-2 count of contiguous blocks, the first one divisible by that count.

Exam trap: "Addresses" = 2^h but "usable hosts" = $2^h - 2$. A /30 has only **2** usable hosts. Subnets = $\mathbf{2^s}$ (old no-subnet-zero rule: $2^s - 2$).

6 IPv6 Addresses

IPv6: (RFC 8200) has **128-bit** addresses: $2^{128} \approx \mathbf{3.4 \times 10^{38}}$ (vast, not unlimited), as **8 groups of 4 hex digits** (16 bits each) separated by colons: 2001:0db8:0000:0000:0000:ff00:0042:8329.



a typical global unicast address: every LAN gets a /64

Abbreviation rules

1. Drop **leading zeros** in each group: 0db8 → db8, 0042 → 42, 0000 → 0
2. Replace **one** run of consecutive all-zero groups by **::** (only once per address; use it for the longest run)
3. Write hex in lowercase (RFC 5952); do not use :: for a single zero group

Compress ✍️:
2001:0db8:0000:0000:0000:ff00:0042:8329
→ 2001:db8:0:0:0:ff00:42:8329
→ **2001:db8::ff00:42:8329**

Expand ✍️: fe80::1:2 shows 3 groups, so :: = 8 − 3 = **5** zero groups → fe80:0:0:0:0:0:1:2 = fe80:0000:0000:0000:0000:0000:0001:0002

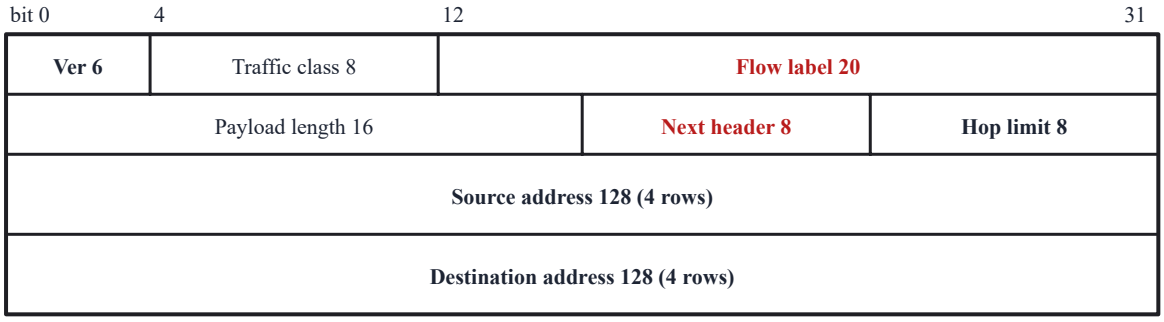
Type	Prefix	Notes
Global unicast	**2000::/3**	public; 2001:db8::/32 = documentation
Link-local	**fe80::/10**	automatic on every interface; never routed
Unique local	fc00::/7 (fd00::/8)	like RFC 1918 private space
Loopback · unspecified	::1/128 · ::/128	like 127.0.0.1 · 0.0.0.0
Multicast	**ff00::/8**	ff02::1 all nodes, ff02::2 all routers
Anycast	(from unicast space)	delivered to the **nearest** member of a group
IPv4-mapped	::ffff:0:0/96	::ffff:192.0.2.1 (dual-stack sockets)

EUI-64 ✍️: MAC 00:1A:2B:3C:4D:5E → split in half and insert **ff:fe** → flip the 7th bit (U/L) of the first byte, 00 → 02 → interface ID **021a:2bff:fe3c:4d5e**, address fe80::21a:2bff:fe3c:4d5e.

Memory trick: "**Double colon, single use**": two :: make the length ambiguous. Prefixes: **2**… global, **fe80** link-local, **ff** multicast.

Exam trap: 2001:db8::1::2 is **invalid** (two ::), so is a group of 5 hex digits. IPv6 has unicast, multicast and anycast, but **no broadcast**.

7 The IPv6 Header (40 bytes, fixed)



8 fields: 64 bits + 2 × 128 bits = 320 bits = 40 bytes

Removed from IPv4: IHL, **header checksum**, identification/flags/offset (moved to the Fragment extension header), options. **Renamed:** TTL → Hop limit, Protocol → Next header, ToS → Traffic class.

Next-header values:
0 Hop-by-Hop · 43 Routing · **44 Fragment**
50 ESP · 51 AH · 60 Destination options
6 TCP · 17 UDP · **58 ICMPv6** · 59 none

Extension headers form a chain: each one's Next header names the one after it, ending in TCP/UDP. Routers normally read only the base header, so forwarding is faster.

Fragmentation is done **only by the source** after Path MTU Discovery; every link must carry at least **1280 bytes**.

Feature	IPv4	IPv6
Address	32-bit, dotted decimal	128-bit, colon hex
Header	20–60 B, 12 fields + options	40 B fixed, 8 fields + extension headers
Checksum / fragments	yes / source + routers	none / source only
Resolution / config	ARP / manual, DHCP	NDP (ICMPv6) / SLAAC, DHCPv6
IPsec	optional add-on	built in (AH/ESP); **use** still optional
QoS / broadcast	DSCP / yes	traffic class + flow label / none

8 IPv4 → IPv6 Transition

- **Dual stack**: devices run IPv4 and IPv6 side by side (most common)
- **Tunnelling**: IPv6 inside IPv4 (protocol **41**): 6in4, **6to4** (2002::/16), **Teredo** (UDP, through NAT), ISATAP
- **Translation**: NAT64 + DNS64 (64:ff9b::/96): IPv6-only hosts reach IPv4 servers



IPv4-only network in between

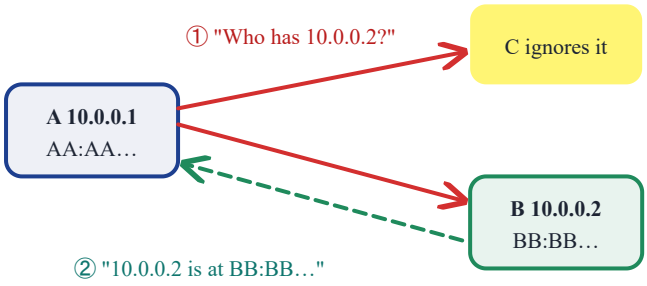
tunnelling = a 6-4-6 sandwich

Memory trick: The three transition roads: "**both, wrap, translate**" = dual stack, tunnel, NAT64.

Exam trap: IPv6 header: **fewer fields** but **bigger** than IPv4's minimum (40 vs 20 bytes).

9 ARP (Address Resolution Protocol)

ARP: (RFC 826, 1982) finds the **MAC address for a known IPv4 address** on the same LAN. Request = **broadcast**, reply = **unicast**; results go in the ARP cache. **EtherType 0x0806**; never crosses a router.



broadcast request, unicast reply

ARP packet: 28 bytes (Ethernet/IPv4):
HW type 2 (1 = Ethernet) · protocol type 2 (0x0800) · HW len 1 (6) · protocol len 1 (4) · **opcode 2**: 1 request, 2 reply (RARP 3, 4) · sender MAC 6, IP 4 · target MAC 6 (zeros in a request), IP 4

- Destination on **another** network → ARP for the **default gateway's MAC**, not the remote host's
- **Gratuitous ARP**: announces its own IP · **Proxy ARP**: a router answers for another subnet
- **ARP spoofing**: forged replies → man-in-the-middle; IPv6 uses NDP instead

10 RARP → BOOTP → DHCP

Protocol	Gives the host	Crosses routers?
RARP (RFC 903, 1984)	MAC → its IP only	no: a server on every LAN (EtherType 0x8035)
BOOTP (RFC 951, 1985)	IP, mask, gateway, boot file	yes, relay agent (UDP 67/68)
DHCP (RFC 2131)	all that + a **lease** (DORA)	yes, relay agent (UDP 67/68)

11 ICMP (Internet Control Message Protocol)

RFC 792; reports errors and answers queries for IP. Carried **inside IP** (protocol **1**) yet counted as network layer. It reports problems but repairs nothing.

- **ping**: Echo request 8 → Echo reply 0
- **traceroute**: probes with TTL 1, 2, 3 … each draw a Time Exceeded from the next router
- No error message about an ICMP error, a broadcast or a non-first fragment

Type	Message
0 / 8	Echo reply / request
3	Dest. unreachable (0 net, 1 host, 3 port, 4 frag. needed + DF)
5	Redirect (a better route)
11	Time exceeded (0 TTL, 1 reassembly)
12	Parameter problem

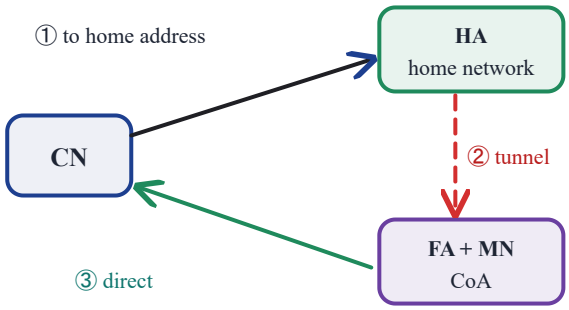
Memory trick: ARP asks "**who has this IP?**", RARP asks "**who am I?**". "**Ping 8, pong 0**; traceroute counts the 11s."

Exam trap: ARP: IP → MAC; **RARP: MAC → IP**. RARP could not cross routers and gave no mask or gateway, so BOOTP and then DHCP replaced it.

12 Mobile IP

Idea: Mobile IP (RFC 5944) lets a mobile node keep its **permanent home address** while it moves, so peers and open connections keep working.

- **Mobile node (MN)**: the roaming host; **correspondent node (CN)**: its peer
- **Home agent (HA)**: router on the home network; keeps the binding, tunnels packets on
- **Foreign agent (FA)**: router on the visited network; offers a CoA, unwraps packets
- **Care-of address (CoA)**: the temporary address; **FA CoA** (the FA's, shared) or **co-located CoA** (from DHCP; the MN unwraps itself)



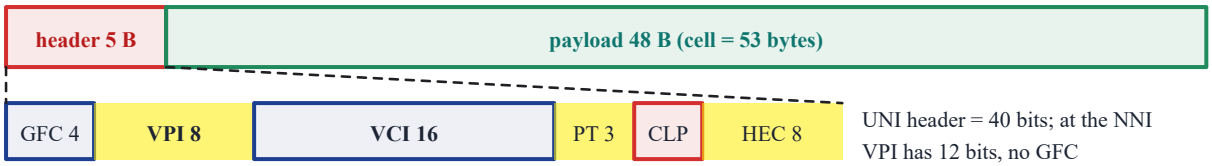
triangle: in via the HA, out direct

1. **Agent discovery**: **agent advertisements** (ICMP router-advertisement extension) or a solicitation
2. **Registration**: MN → FA → HA on **UDP 434**; the HA binds home address ↔ CoA
3. **Tunnelling**: the HA intercepts (proxy ARP) and sends **IP-in-IP** (protocol 4) to the CoA

Exam trap: Replies go MN → CN directly, so traffic makes a **triangle** ("triangle routing"). Fixes: route optimisation (the CN learns the CoA), reverse tunnelling. **Mobile IPv6** (RFC 6275) has **no foreign agent**.

13 Network Layer of ATM

Idea: **Connection-oriented, cell-switched**: a **virtual circuit** is set up once, then 53-byte cells are switched by **VPI/VCI labels** under the agreed QoS; cells stay in order.



- VPI: 256 paths at the UNI, **4,096** at the NNI; VCI: 65,536 per path
- **VP switch** rewrites only the VPI; **VC switch** rewrites VPI and VCI; labels are **local** to each link
- **PVC** (configured) or **SVC** (signalled, Q.2931)
- ATM addresses (20-byte NSAP or E.164) are used only at set-up

Service class	Typical use
CBR	voice, circuit emulation
rt-/nrt-VBR	video / bursty data
ABR	RM-cell feedback
UBR	best effort

Policing: GCRA (leaky bucket); CLP = 1 cells are dropped first.

Memory trick: **"Path is the motorway, channel is the lane"**: the VPI picks the bundle, the VCI the circuit inside it. HEC = CRC-8 over the header only; efficiency 48/53 ≈ 90.6%. Cell delineation: ATM data-link notes; AAL: transport notes.

Night-Before-the-Exam Sheet

Fact sheet:

IPv4 header 20–60 B · IHL × 4	Total length max 65,535 B
Protocol: 1 ICMP, 6 TCP, 17 UDP	TTL 0 → drop + ICMP type 11
A 1–126 · B 128–191 · C 192–223	D 224–239 multicast · E 240–255
10/8 · 172.16/12 · 192.168/16 private	127/8 loopback · 169.254/16 APIPA
hosts = $2^h - 2$ · block = 256 − mask	/26 = .192, 62 hosts · /30 = 2 hosts
IPv6 128-bit, header 40 B, 8 fields	:: only once · no broadcast
fe80::/10 link-local · ff00::/8 multicast	2000::/3 global · ::1 loopback
ARP IP → MAC, 0x0806, 28 B	RARP MAC → IP → BOOTP → DHCP
Mobile IP: HA, FA, CoA, UDP 434	ATM 53 = 5 + 48 · VPI 8/12, VCI 16

Top 14 one-liners for MCQs

1. Proxy = application-level gateway (layer 7)
2. Stateful firewall keeps a state table
3. DMZ = screened subnet for public servers
4. IPv4 checksum covers the header only
5. DF = don't fragment, MF = more fragments
8. IPv6 = $2^{128} \approx 3.4 \times 10^{38}$ addresses
9. IPv6 fragments at the source only; min MTU 1280
10. NDP (ICMPv6) replaces ARP in IPv6
11. Dual stack, tunnelling, NAT64
12. ping = ICMP echo 8 / 0

6. PAT maps many hosts to one IP by port

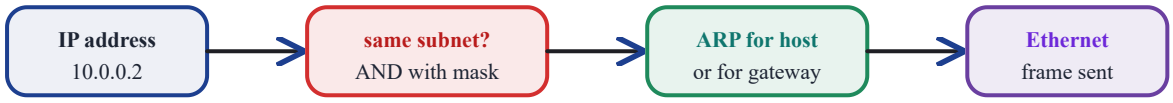
7. 200.10.0.0–3.0/24 aggregate to /22

13. MIPv4 tunnel: IP-in-IP, HA → CoA

14. ATM HEC protects the 5-byte header only

Top traps: 172.32.x.x is public · usable hosts = $2^h - 2$, not 2^h · /30 = 2 hosts · two :: are invalid · IPv6 has no broadcast and no header checksum · IPsec is designed into IPv6 but its use is optional · NAT is not a firewall · ARP reply is unicast · VPI/VCI are local labels, not addresses

All tricks together: Envelope / conversation / letter (filter / stateful / proxy) · IHL × 4, offset × 8 · "1 ping, 6 TCP, 17 UDP" · magic number = 256 − mask · double colon, single use · both, wrap, translate · ping 8, pong 0 · path = motorway, channel = lane



how a host sends one IPv4 packet: mask, ARP, frame

✓ Practice MCQs (exam pattern, 10 marks)

1. A packet filter decides mainly on:
(a) application data (b) IP, port, protocol
(c) the user's login (d) MAC address only

2. An IPv4 header with IHL = 6 is how long?
(a) 6 bytes (b) 12 bytes
(c) 24 bytes (d) 48 bytes

3. Which address is private (RFC 1918)?
(a) 172.31.200.1 (b) 172.32.0.1
(c) 192.169.0.1 (d) 11.0.0.1

4. Usable host addresses in a /28 subnet:
(a) 16 (b) 30
(c) 6 (d) 14

5. Broadcast address of 192.168.10.130/26:
(a) 192.168.10.127 (b) 192.168.10.255
(c) 192.168.10.192 (d) 192.168.10.191

6. 2001:0db8:0:0:0:0:0:0001 compresses to:
(a) 2001:db8::1 (b) 2001:db8:::1
(c) 2001::db8::1 (d) 21:db8::1

7. Size of the IPv6 base header:
(a) 20 bytes (b) 60 bytes
(c) 40 bytes (d) 128 bytes

8. RARP maps:
(a) IP → MAC (b) MAC → IP
(c) name → IP (d) IP → name

9. In Mobile IP, the temporary address of a mobile node in a foreign network is its:
(a) home address (b) MAC address
(c) loopback address (d) care-of address

10. Size of an ATM cell header:
(a) 48 bytes (b) 53 bytes
(c) 5 bytes (d) 8 bytes

Answer key → 1-b · 2-c · 3-a · 4-d · 5-d · 6-a · 7-c · 8-b · 9-d · 10-c

Written question: The network 192.168.5.0/24 must be divided into 4 equal subnets. Find the new mask, the usable hosts per subnet, and for each subnet its network address, first and last usable host and broadcast address.

8 marks

1. $4 = 2^2$ → borrow **2** bits → **/26** (255.255.255.192) 2
2. h = 32 − 26 = 6 → 64 addresses, **62 usable**; block 256 − 192 = 64 2
3. Subnets start at multiples of 64 (table) 4

Network	First	Last	Bcast
192.168.5.0	.1	.62	.63
192.168.5.64	.65	.126	.127
192.168.5.128	.129	.190	.191
192.168.5.192	.193	.254	.255

Before the exam ✓

- ✓ Firewall types and a DMZ diagram
- ✓ IPv4 / IPv6 headers with field sizes
- ✓ Network, broadcast, hosts for any /n
- ✓ Compress / expand an IPv6 address

Also practise (written):

① Mobile IP with a diagram
② ARP request/reply steps
③ VLSM for 100, 50, 2 hosts
④ IPv4 vs IPv6 (8 points)